

PROGRAM ASSESSMENT REPORT

v2024.1

Tracking data

CIM code	MAS-ACDF1
Program name	Masters of Science in Applied Cybersecurity and Digital Forensics
year	2024
Program assessment coordinator	Marwan Omar
Assessment plan year	2024-2025
No assessment this year per plan	☐
LO(s) assessed	1,3
Presented at program authority meeting date: <i>Please attach minutes</i>	

Learning Outcomes assessed in ITMS 548 for Academic Year 2024-2025

LO 1	Design and implement a comprehensive enterprise security program using both policy and technology to implement technical, operational, and managerial controls.
LO 3	Technically secure enterprise information assets and resources to deter, detect, and prevent the success of attacks and intrusions.

Methodology

For each LO and artifact assessed, please describe methodology used, and attach (as supplemental files during report submission) or insert here the rubric used. At minimum this should contain:

- a description of how the artifact is concordant with the LO
- a rubric describing the method used to place individual student in at least 3 assessment categories; {**D** Does not meet expectations, **M** Meets expectations *M*, **P** Proficient}

Learning Objective 2: Navigating Linux

Description:

This learning objective focuses on assessing students' proficiency in navigating and utilizing Linux operating systems. Linux is a fundamental operating system used in many cybersecurity and IT environments due to its flexibility, security, and powerful command-line tools. Understanding Linux is crucial for performing tasks such as file management, system administration, network configuration, and scripting.

Student Performance Analysis:

Students' performance on this objective shows their ability to work with the Linux operating system, manage files, execute commands, and write basic scripts. High-performing students (categorized as "P") have demonstrated a strong command over

these skills, indicating readiness to handle real-world IT or cybersecurity tasks that require Linux. Students categorized as "M" have an adequate understanding but may need additional practice. Students categorized as "D" need significant improvement, as proficiency in Linux is essential for many technical roles.

Learning Objective 5: Physical and Environmental Security

Description:

This learning objective measures students' understanding of physical and environmental security concepts. It includes knowledge about securing physical environments, understanding threats like unauthorized access, environmental hazards, and implementing measures to protect sensitive information from these threats.

Student Performance Analysis:

Performance in this area shows how well students can apply physical security principles, such as implementing access controls, securing facilities, and understanding environmental threats. A high score (P) indicates that students can effectively identify vulnerabilities and suggest measures to mitigate risks, which is essential for roles in security management and infrastructure protection. Lower scores (M or D) suggest a need for deeper comprehension of security protocols and risk management practices.

Learning Objective 7: Exploring the Fundamentals of Cryptography

Description:

This learning objective evaluates students' knowledge and practical understanding of cryptographic principles. Cryptography is a core component of cybersecurity, focusing on techniques such as encryption, decryption, hashing, and digital signatures to protect data integrity and confidentiality.

Student Performance Analysis:

Students who achieve a "P" in this category have shown a solid grasp of cryptographic concepts and their applications, including how to encrypt/decrypt data and secure communications. They are well-prepared for tasks that require data protection and secure communication protocols. Scores in the "M" or "D" range indicate that students may struggle with understanding complex cryptographic techniques and require further study to gain proficiency in this critical area of cybersecurity.

Spring 2025 Analysis

Learning Objective 2: Navigating Linux

Description:

This learning objective focuses on equipping students with the skills necessary to

navigate and utilize Linux operating systems effectively. Linux is a widely used operating system in IT and cybersecurity environments due to its robustness, flexibility, and the level of control it offers over system resources. Students are expected to demonstrate proficiency in executing commands, managing files, directories, users, and permissions, and understanding shell scripting to automate tasks.

Significance in the Curriculum:

Mastery of Linux is crucial for cybersecurity professionals, as many security tools and servers run on Linux platforms. This objective prepares students for roles where they will need to use Linux for tasks like network monitoring, system administration, and penetration testing. Students performing well in this objective are likely to excel in technical roles where Linux proficiency is a requirement.

What Student Performance Indicates:

- **Proficient (P):** Students with a high grade have a strong grasp of Linux commands, file management, and scripting. They can navigate the Linux environment confidently and execute complex tasks, showing readiness for real-world technical challenges.
- **Moderate (M):** Students with moderate scores understand basic Linux operations but may struggle with more advanced topics or scripting. Additional practice will help them achieve greater command over the operating system.
- **Developing (D):** Students need significant improvement in using Linux. This lack of proficiency may limit their effectiveness in technical roles that require operating system management or security tasks.

Learning Objective 5: Physical and Environmental Security**Description:**

This learning objective evaluates students' understanding of physical and environmental security measures. Physical security involves protecting facilities, hardware, and other physical assets from unauthorized access, damage, or theft. Environmental security addresses risks posed by environmental factors such as natural disasters, temperature fluctuations, and humidity, which can impact IT infrastructure and sensitive data.

Significance in the Curriculum:

Understanding and implementing physical and environmental security measures is fundamental to ensuring the overall safety and functionality of IT systems. This objective prepares students to assess and manage security risks associated with physical access to facilities and environmental controls. It is a key component for roles in security management, facility oversight, and disaster recovery planning.

What Student Performance Indicates:

- **Proficient (P):** High-scoring students have demonstrated a solid understanding of physical and environmental security principles. They can effectively apply concepts to safeguard assets, implement access controls, and design secure facilities.

- **Moderate (M):** Students with moderate scores may understand basic principles but lack depth in applying advanced security measures or analyzing environmental risks. They may need additional training in risk assessment and mitigation strategies.
- **Developing (D):** Students need a better grasp of physical and environmental security concepts. They may not yet be able to identify threats or recommend adequate controls to protect physical assets, limiting their capability to contribute to security planning.

Learning Objective 7: Exploring the Fundamentals of Cryptography

Description:

This learning objective focuses on the theoretical and practical aspects of cryptography. Cryptography is essential for securing digital communications and protecting data integrity and confidentiality. Students are expected to understand encryption and decryption processes, cryptographic algorithms, key management, and digital signatures.

Significance in the Curriculum:

Cryptography is a cornerstone of cybersecurity, underpinning many security protocols and standards. This objective ensures that students gain foundational knowledge in cryptographic techniques and their applications, preparing them for tasks such as secure communication, data encryption, and authentication processes.

What Student Performance Indicates:

- **Proficient (P):** Students with a high score have a strong grasp of cryptographic principles and can apply them to secure data and communications. They understand how to implement cryptographic solutions and evaluate their effectiveness, indicating readiness for roles involving data security and cryptography.
- **Moderate (M):** Students with moderate scores have a basic understanding but may struggle with complex concepts like asymmetric encryption or key management. Further study and practice are needed to strengthen their command over cryptographic solutions.
- **Developing (D):** Students lack a foundational understanding of cryptographic principles. This deficiency may hinder their ability to work with encryption technologies, limiting their effectiveness in security-related roles that require data protection skills.

Results

Artifact samples

Attach artifact samples. For data sets with ≤ 10 students, provide 1 in each category (unless the category is empty, if so state that). For programs with ≤ 20 , provide 2. For other programs provide 3. Please redact any personally identifiable information, PII (which included names and A#s).

[Artifacts link](#)

Data table

For each LO and provide a table pasted in from the [provided template](#) placing each student into the assessment categories (minimally {D, M, P}). Please redact any personally identifiable information, PII (including names or A#s) from this report. Provide a summary graph from the template presenting this distribution. You should also present additional analysis as deemed appropriate by the results seen, and the needs of the program.

Description of findings

Summarize the discussion that occurred within the program authority when this was presented. Provide a short narrative interpreting the findings. Are there any areas of concern?

Description of Findings

The review of the **ITMS 548: Cybersecurity Technologies course** focused on the key learning objectives: **Navigating Linux**, **Physical and Environmental Security**, and **Exploring the Fundamentals of Cryptography**. After evaluating the student performance and applying the rubric, the following findings were observed.

Key Findings:

1. Learning Objective 2: Navigating Linux

- **Proficient (P):** The vast majority of students demonstrated excellent proficiency in navigating Linux environments. They were able to confidently execute advanced commands, manage files and directories, and write effective shell scripts to automate tasks. These students are well-prepared for technical roles where Linux is a core requirement, such as system administration and cybersecurity operations.
- **Moderate (M):** Only a small number of students required further practice in more complex Linux scripting and permission management, but they displayed a solid foundational understanding.

2. Learning Objective 5: Physical and Environmental Security

- **Proficient (P):** Most students performed very well in this area, showing a solid understanding of **physical security controls** and how to mitigate **environmental risks**. They effectively demonstrated how to secure facilities, implement access controls, and protect IT assets from environmental hazards, preparing them for roles in **security management**.
- **Moderate (M):** A small minority of students had minor gaps in identifying environmental threats or fully implementing comprehensive physical security controls, but their overall understanding was still adequate.

3. Learning Objective 7: Exploring the Fundamentals of Cryptography

- **Proficient (P):** The vast majority of students excelled in cryptographic concepts, particularly in **encryption**, **decryption**, and **key management**. They were able to apply

these principles effectively in securing data and communications, showcasing their readiness for advanced cybersecurity roles.

- **Moderate (M):** Only a small group struggled with more complex cryptographic techniques such as **asymmetric encryption** and **digital signatures**, indicating a need for additional focus on these areas in future coursework.

Interpretation of Findings:

Overall, the student cohort performed exceptionally well in all three learning objectives, with most students categorized as **Proficient (P)**. This reflects strong comprehension and application of key cybersecurity concepts such as Linux navigation, physical security, and cryptographic techniques. The **proficiency** demonstrated by students suggests they are well-prepared for technical roles in cybersecurity, where these skills are critical.

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Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, and if areas for possible improvement emerge, describe plans to achieve this improvement.

Recommendations:

- While the majority of students performed at a high level, there is a minor suggestion for improvement: offering **optional advanced workshops** on more complex cryptographic techniques, such as **asymmetric encryption and digital signatures**, could benefit students who struggled with these areas and enhance overall course comprehension.

This minor adjustment will help ensure all students are equipped with a deeper understanding of **advanced encryption methods**, further preparing them for real-world cybersecurity challenges

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)? no previous CI plans available because this is the first time we ran this course.

This cycle's CI plan

Describe any CI plans developed from this cycle's data, as well as the strategy developed to implement these plans.

☐ After consideration and presentation to the program authority, no CI plans are recommended.

- While the majority of students performed at a high level, there is a minor suggestion for improvement: offering **optional advanced workshops** on more complex cryptographic techniques, such as **asymmetric encryption and digital signatures**, could benefit students who struggled with these areas and enhance overall course comprehension.

This minor adjustment will help ensure all students are equipped with a deeper understanding of **advanced encryption methods**, further preparing them for real-world cybersecurity challenges